UNIVERSITY OF PETROLEUM AND ENERGY STUDIES

IC Fabrication and Packaging

PBL 1 — Interconnect Resistance, Capacitance & RC Behavior

Submission details	
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Abstract

This report examines package interconnect resistance, capacitance and signal integrity using Ansys 2D Extractor and Circuit. Organic/Thin Film microstrip, stripline and embedded microstrip structures are compared through geometric and dielectric sweeps. Coupled ceramic interconnects are evaluated for loss and capacitive coupling, while PWB soldermask effects are assessed with and without dielectric coverage. Finally, a 15 mm coupled embedded-microstrip RC model is used to estimate receiver delay, rise time and quiet-line crosstalk. All quantitative findings below correspond to the supplied simulation evidence, with modeling limitations stated explicitly.

1. Simulation methodology

Parameterized 2D cross-sections were constructed with copper conductors (conductivity 5.8×10^7 S/m). Capacitance/conductance (CG) and resistance/inductance (RL) were extracted using AEDT Q2D, principally at 1 GHz. The ground/substrate width was nominally 2000 µm. The capacitance matrix follows the Maxwell convention, so the off-diagonal term C12 is negative; the equivalent coupling capacitor is |C12|. For transient modeling, per-unit-length values were multiplied by 0.015 m.

Substrate	εr	tan δ	W (µm)	S (µm)	T (µm)	H (µm)
PWB	4.4	0.020	150	280	35	180
Ceramic	9.8	0.002	80	300	15	125
Organic/Thin Film	3.2	0.010	20	30	5	12

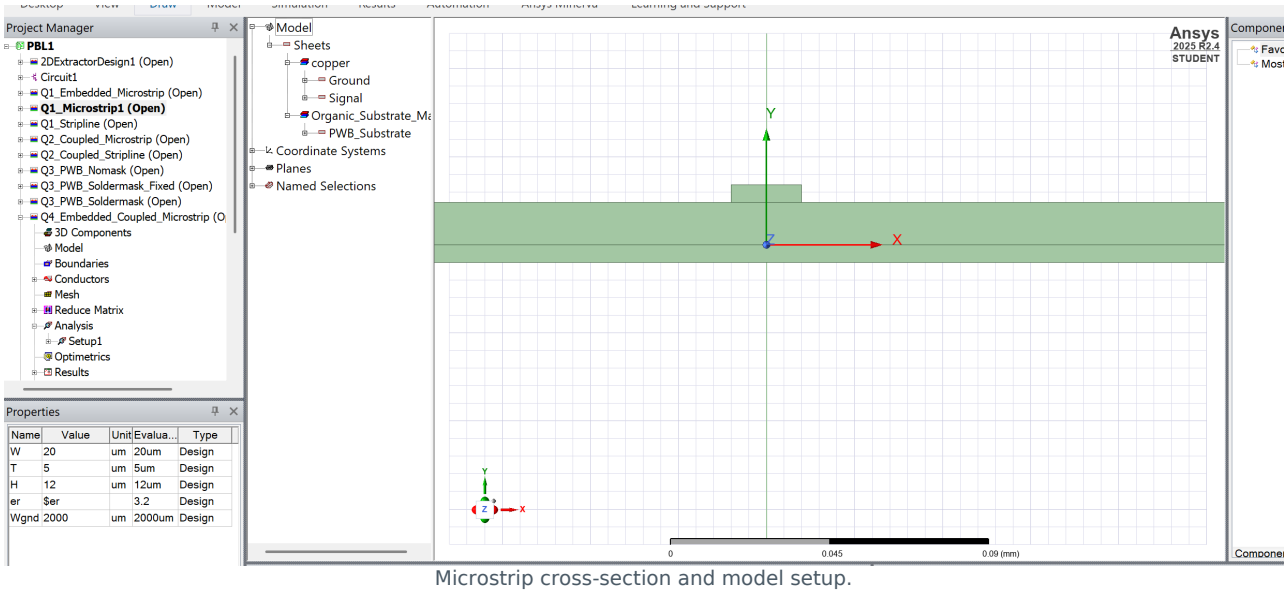
These technology dimensions follow the supplied ECEG4071 PBL 1 brief. Embedded models additionally use an assumed 6 µm dielectric cover, documented as a modeling choice.

2. Q1 — Geometry and dielectric sensitivity

The Q1 base case uses Organic/Thin Film material. For each topology, width W was varied from 10 to 40 µm, signal-to-ground spacing H from 6 to 24 µm, relative permittivity from 2.5 to 4.5, and ground-plane width from 100 to 2000 µm. Increasing W and εr generally increased capacitance, while increasing H decreased it. Ground-width plots showed much smaller variations. The ground-width threshold has not been independently established to a precise sub-1% cutoff; these plots are retained as qualitative evidence.

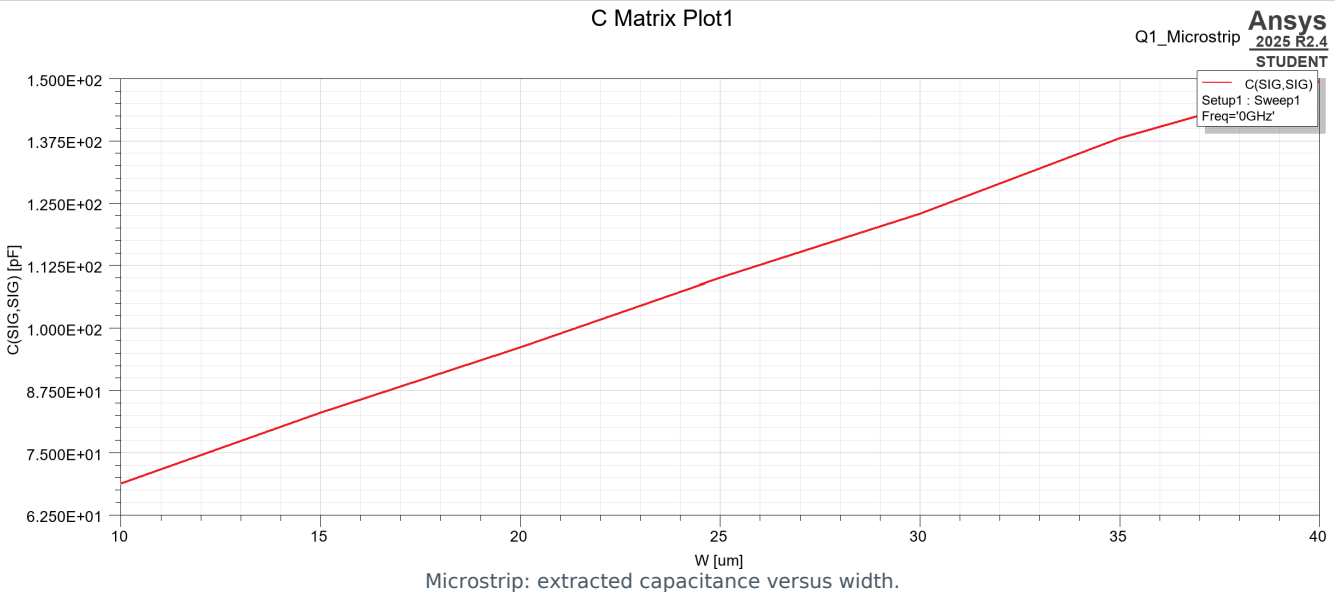
The three structures exhibit different dielectric field distributions: microstrip has an air-facing field component, embedded microstrip adds dielectric above the trace, and stripline is bounded by reference planes. Numerical values for each sweep are retained in the original AEDT plots rather than inferred from the plotted curves.

Microstrip — geometry

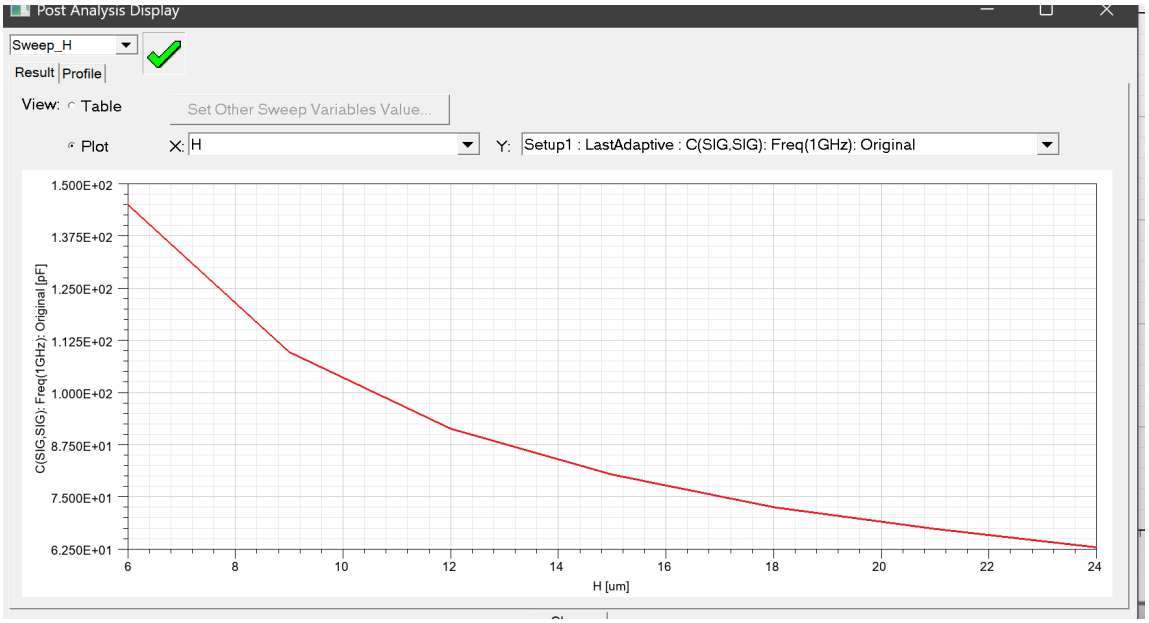


Microstrip cross-section and model setup.

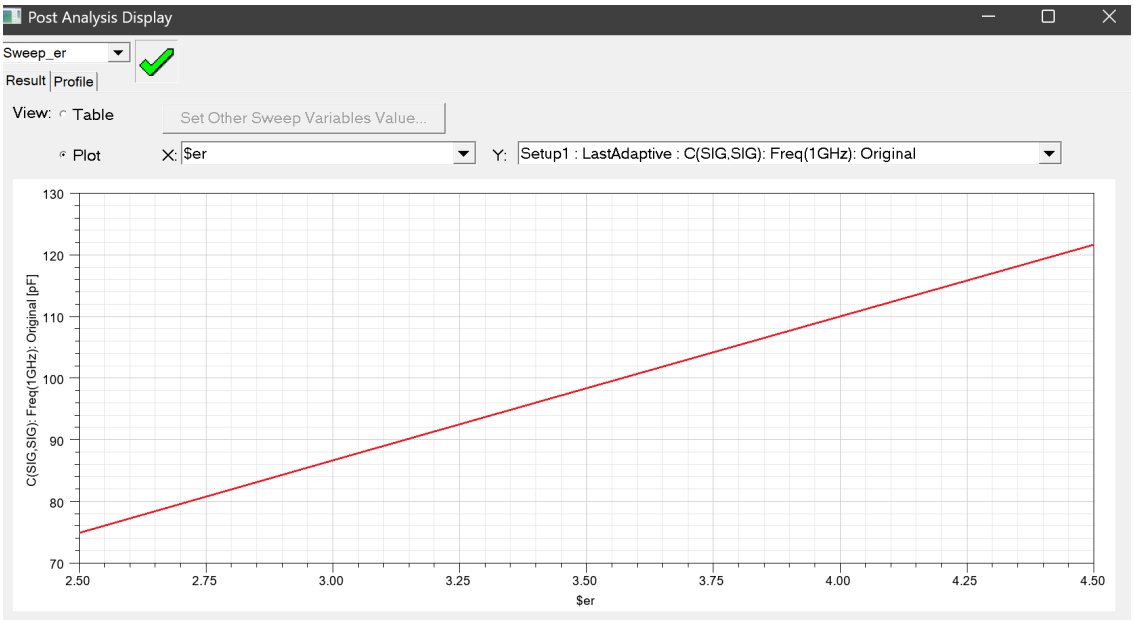
Microstrip — sweep plots



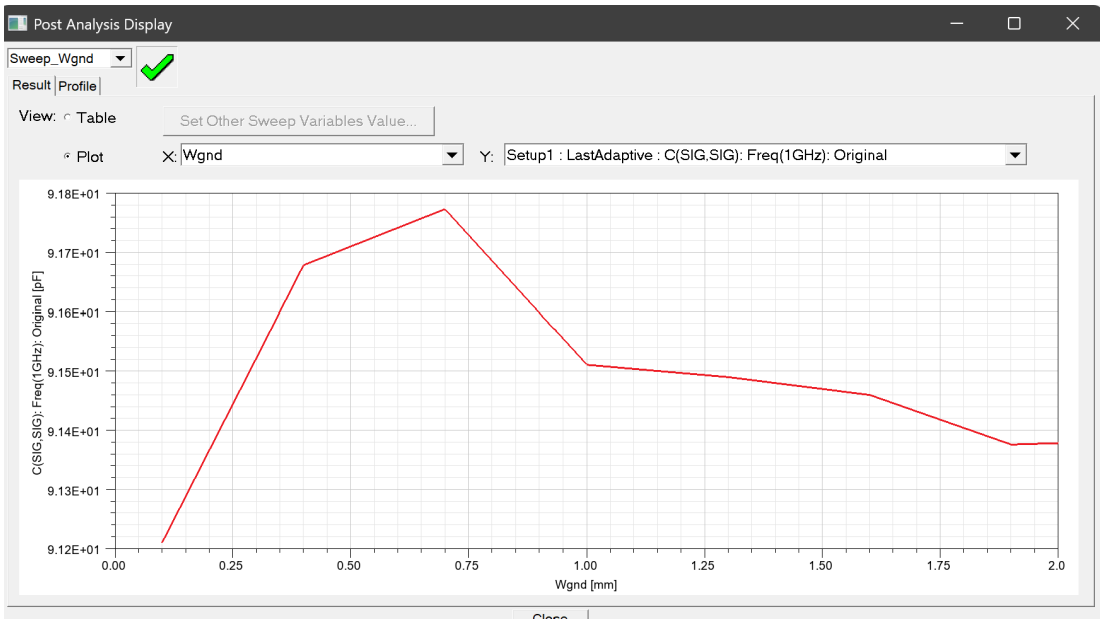
Microstrip: extracted capacitance versus width.



Microstrip: extracted capacitance versus height.

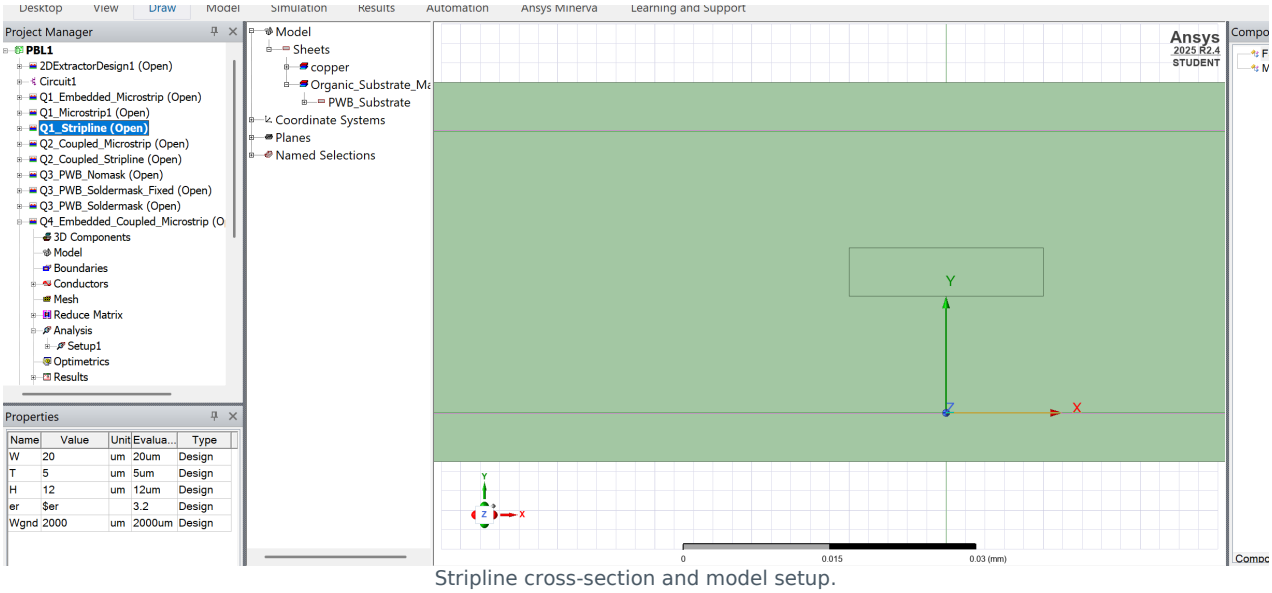


Microstrip: extracted capacitance versus relative permittivity.



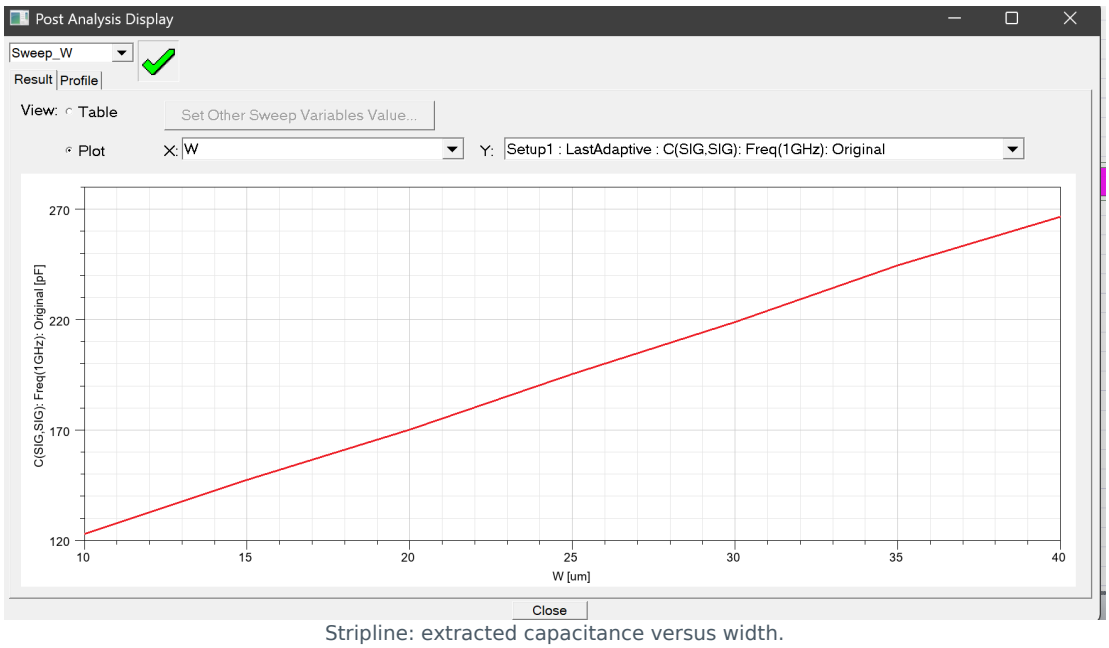
Microstrip: extracted capacitance versus ground width.

Stripline — geometry

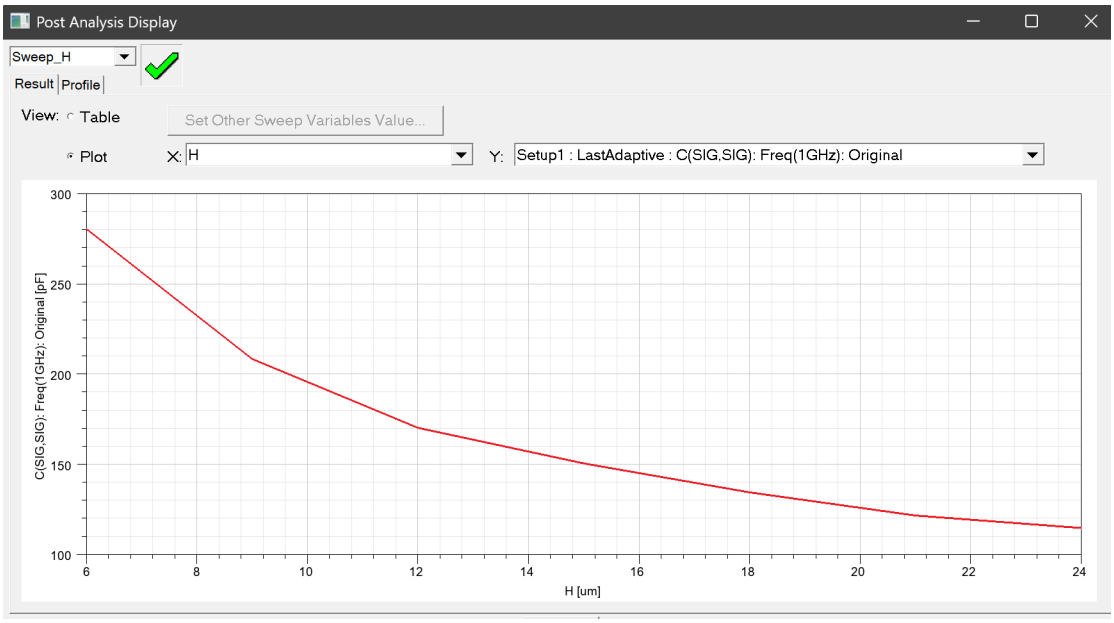


Stripline cross-section and model setup.

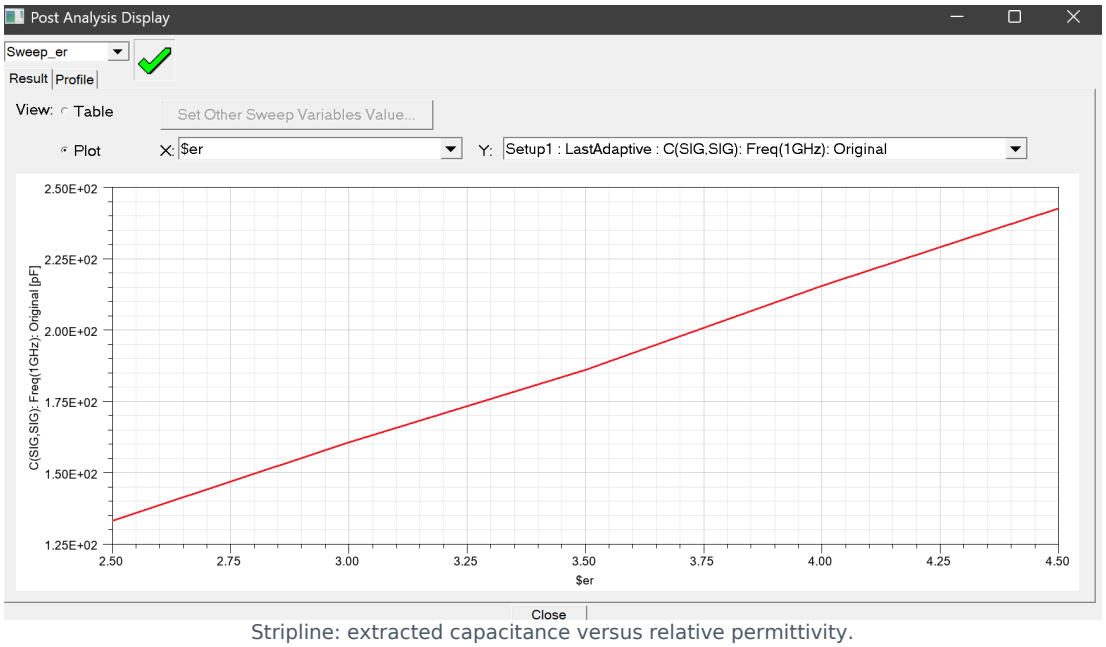
Stripline — sweep plots



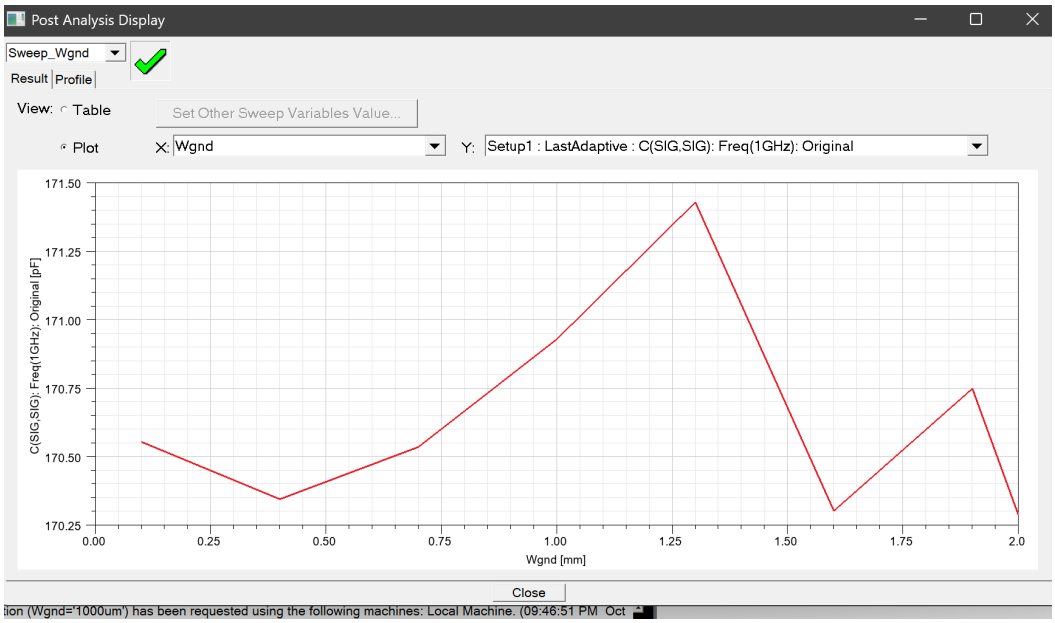
Stripline: extracted capacitance versus width.



Stripline: extracted capacitance versus height.

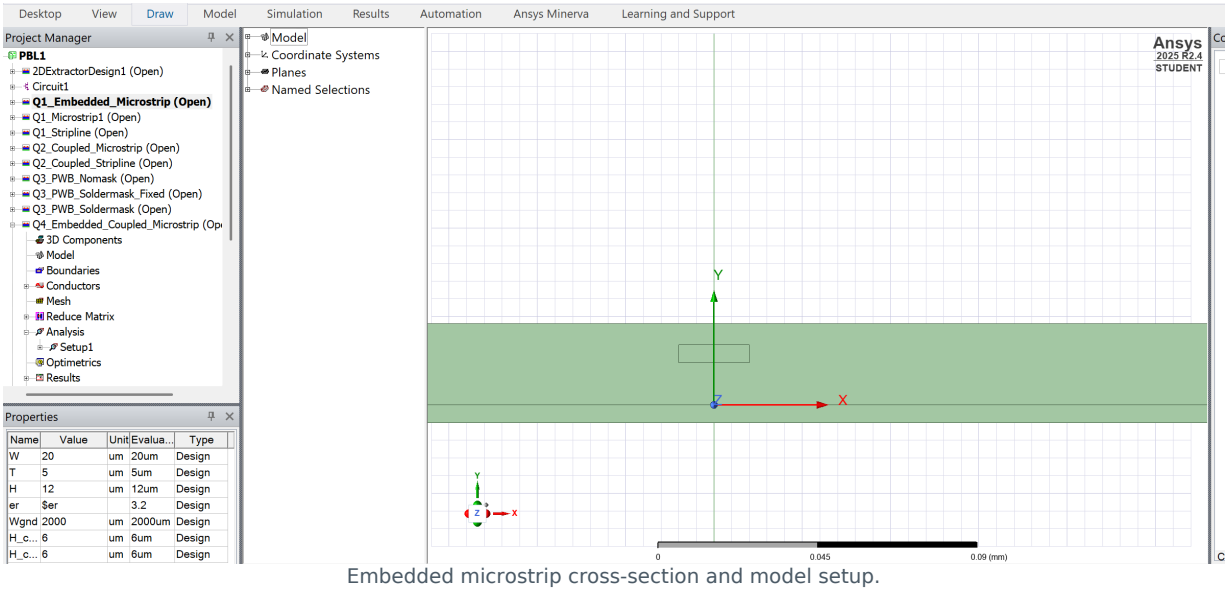


Stripline: extracted capacitance versus relative permittivity.



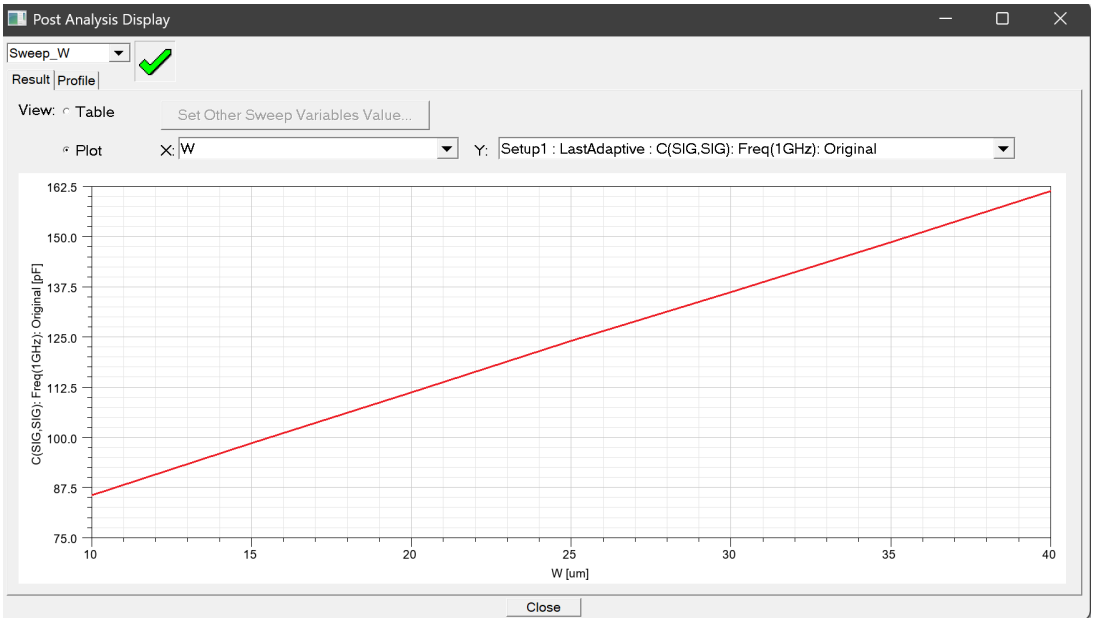
Stripline: extracted capacitance versus ground width.

Embedded microstrip — geometry

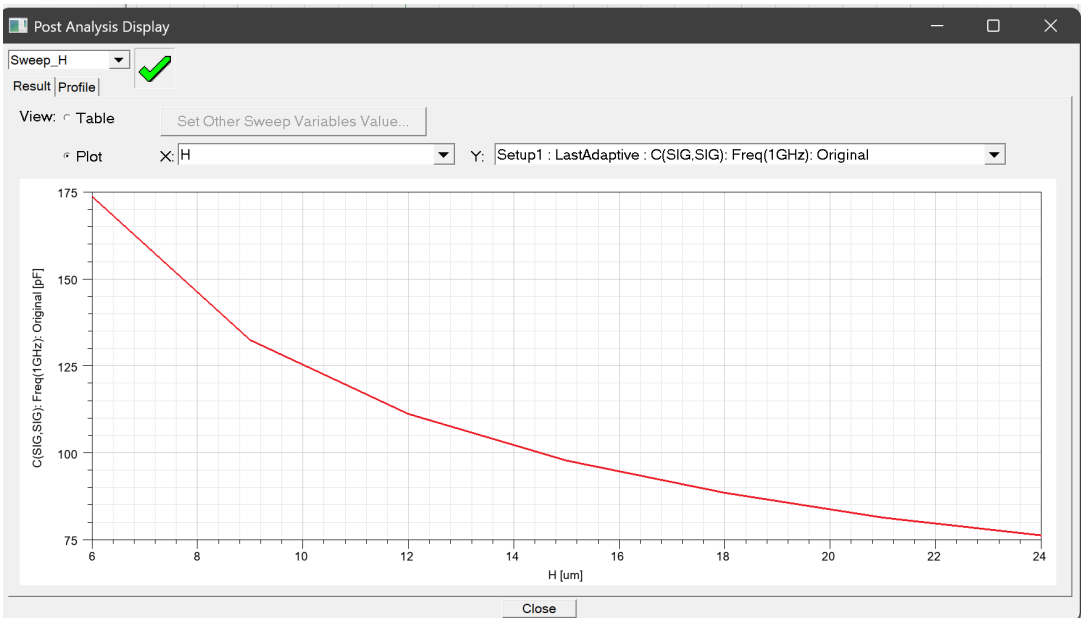


Embedded microstrip cross-section and model setup.

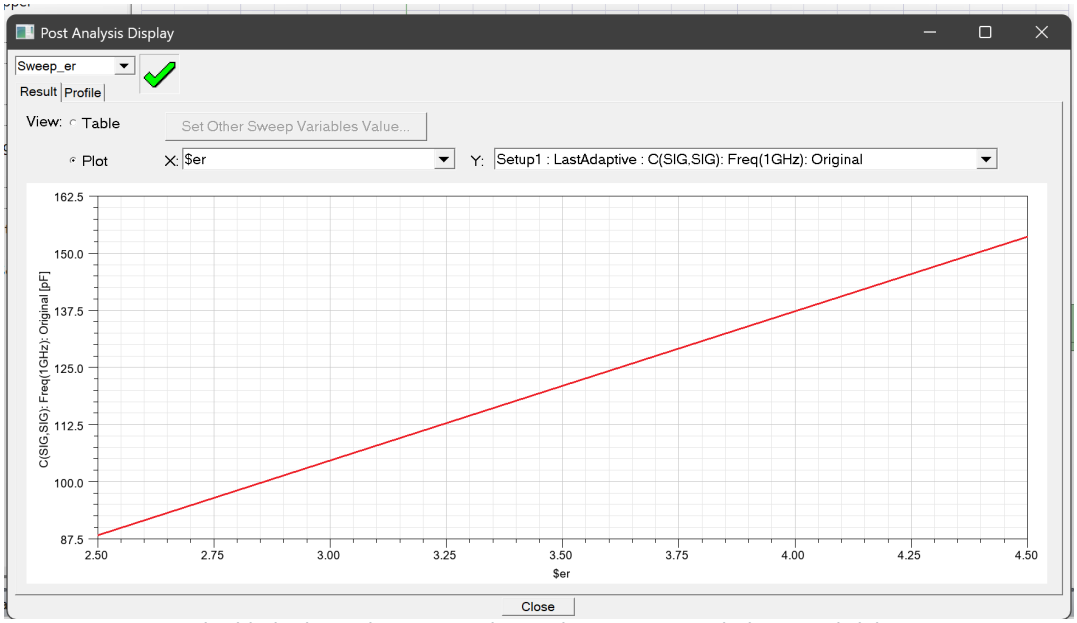
Embedded microstrip — sweep plots



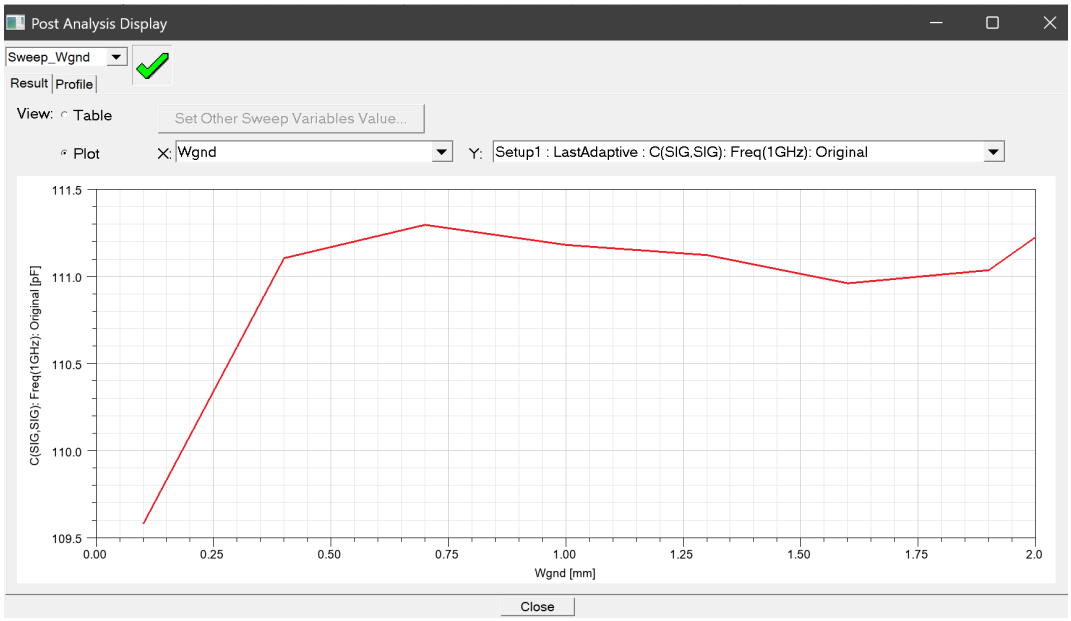
Embedded microstrip: extracted capacitance versus width.



Embedded microstrip: extracted capacitance versus height.



Embedded microstrip: extracted capacitance versus relative permittivity.



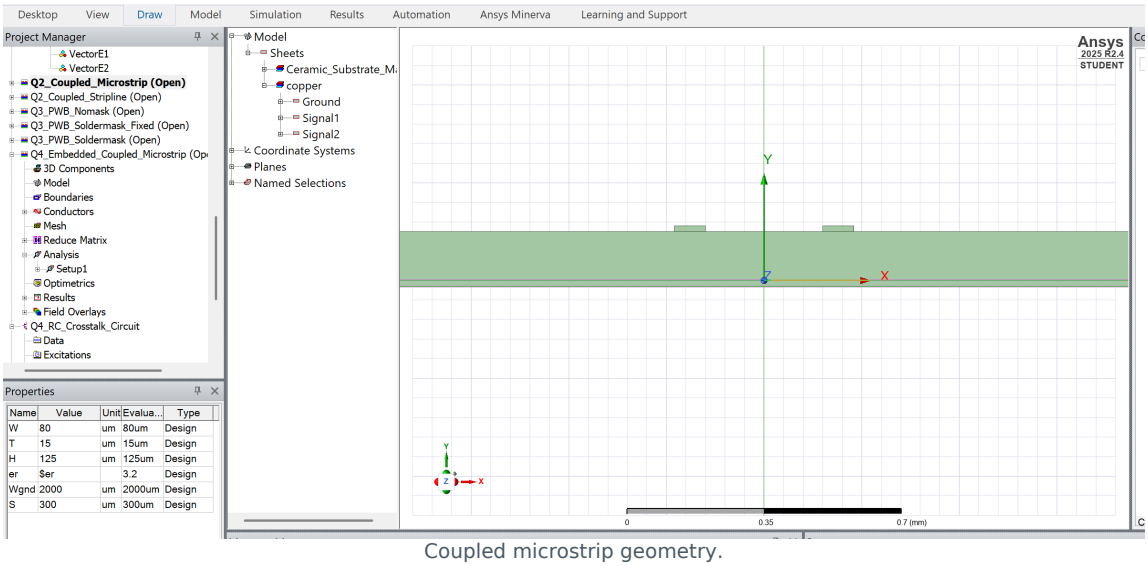
Embedded microstrip: extracted capacitance versus ground width.

3. Q2 — Coupled ceramic microstrip and stripline

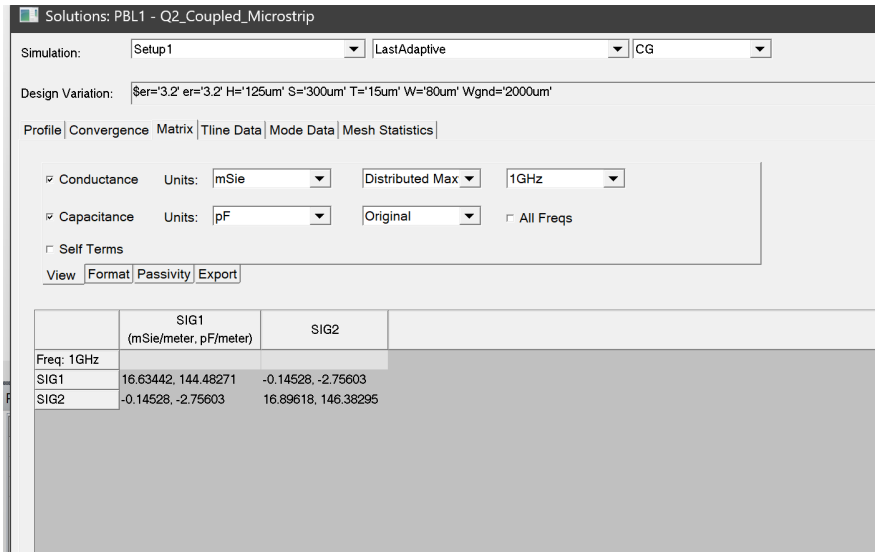
Two 80 µm-wide ceramic traces were separated by 300 µm edge-to-edge. The extracted 1 GHz matrices demonstrate higher absolute mutual capacitance for the stripline, but lower normalized coupling. Modal attenuation is lower for the microstrip in both modes, making it preferable for the modeled signal-loss objective.

Quantity at 1 GHz	Coupled microstrip	Coupled stripline
C11 (pF/m)	144.48271	293.13472
C12 (pF/m)	−2.75603	−3.98072
|C12|/C11	1.91%	1.36%
R11 (Ω/m)	78.21584	76.93484
Re(Z0,11) (Ω)	57.205	36.167
Mode 1 attenuation (dB/m)	10.28	14.81
Mode 2 attenuation (dB/m)	9.92	15.02

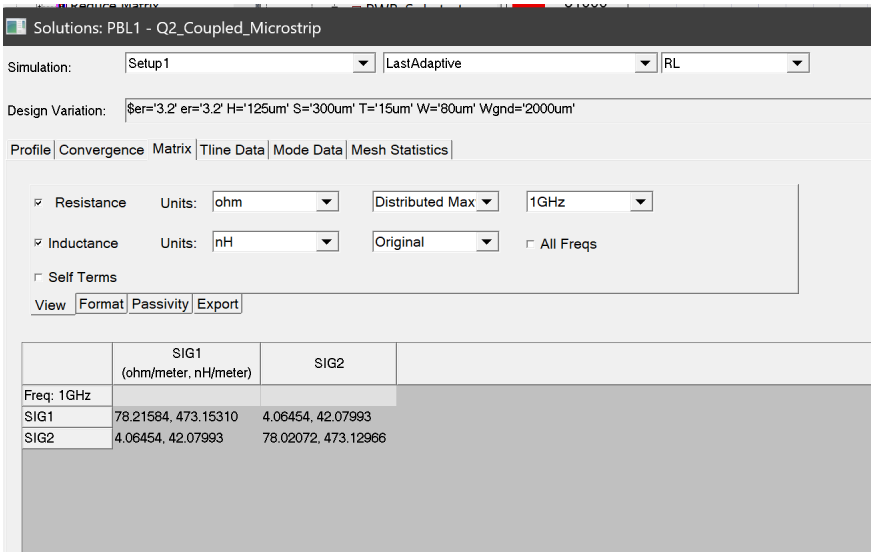
Q2(a), loss: Although the stripline has slightly lower conductor resistance, its extracted modal attenuation is higher; the ceramic microstrip is preferable for minimizing total attenuation under these simulated conditions. Q2(b), capacitive coupling: Stripline reduces the normalized ratio |C12|/C11 from 1.91% to 1.36%, although its absolute |C12| increases. This ratio alone does not establish the quiet-line voltage for arbitrary terminations.



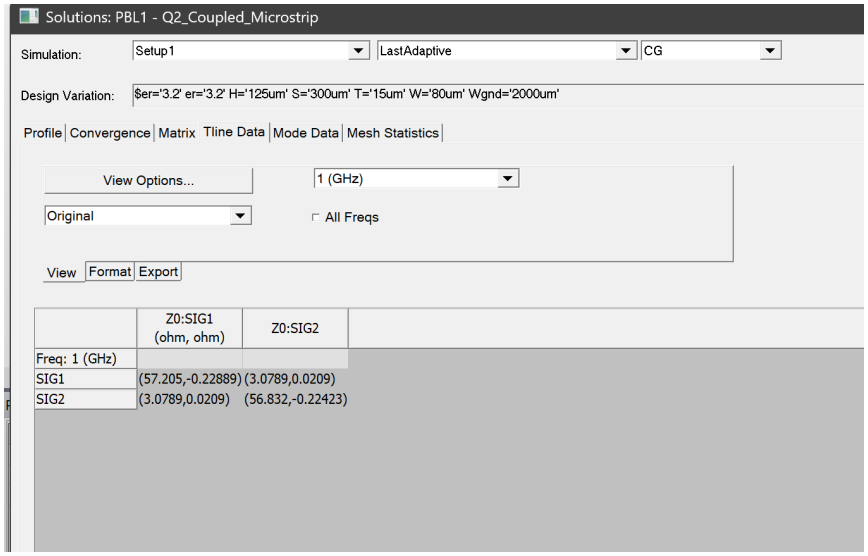
Coupled microstrip geometry.



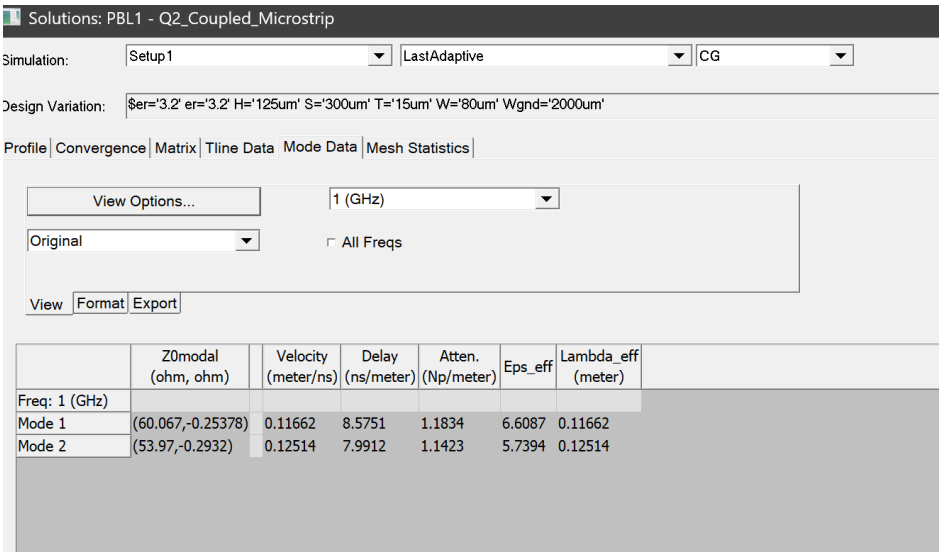
Coupled microstrip — Capacitance/conductance matrix.



Coupled microstrip — Resistance/inductance matrix.

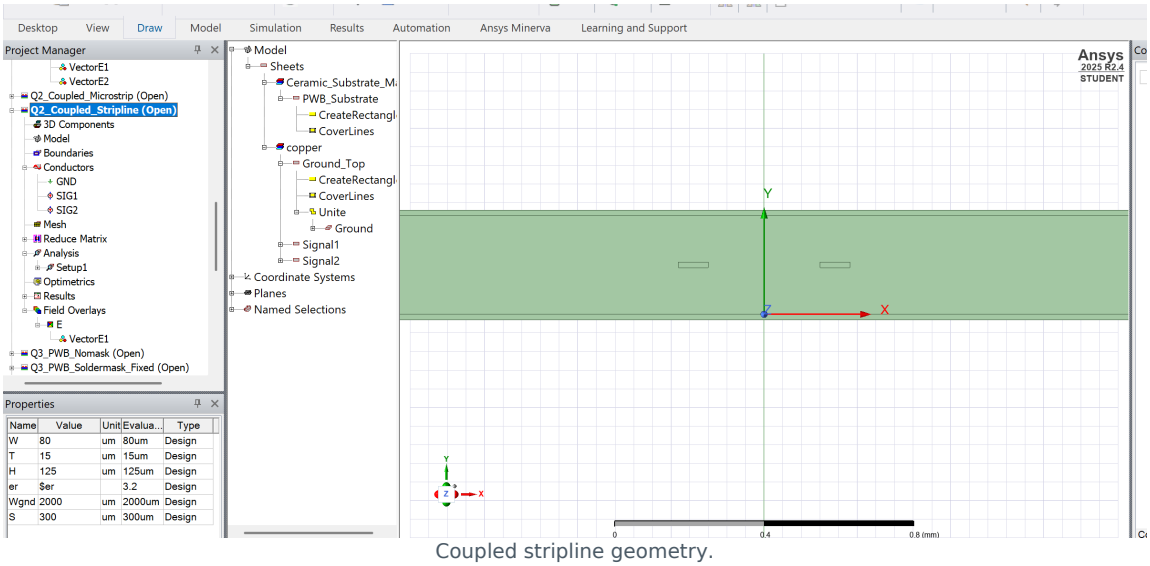


Coupled microstrip — Transmission-line impedance data.

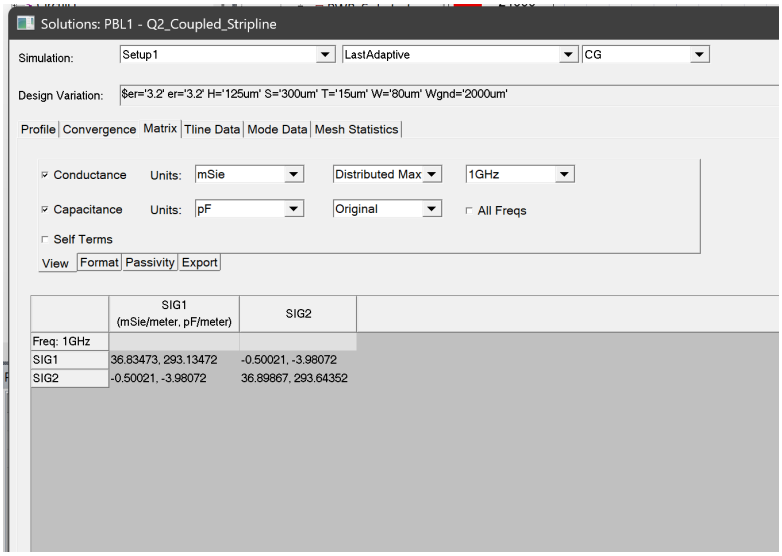


	Z0modal (ohm, ohm)		Velocity (meter/ns)	Delay (ns/meter)	Atten. (Np/meter)	Eps_eff	Lambda_eff (meter)
Freq: 1 (GHz)							
Mode 1	(60.067,-0.25378)		0.11662	8.5751	1.1834	6.6087	0.11662
Mode 2	(53.97,-0.2932)		0.12514	7.9912	1.1423	5.7394	0.12514

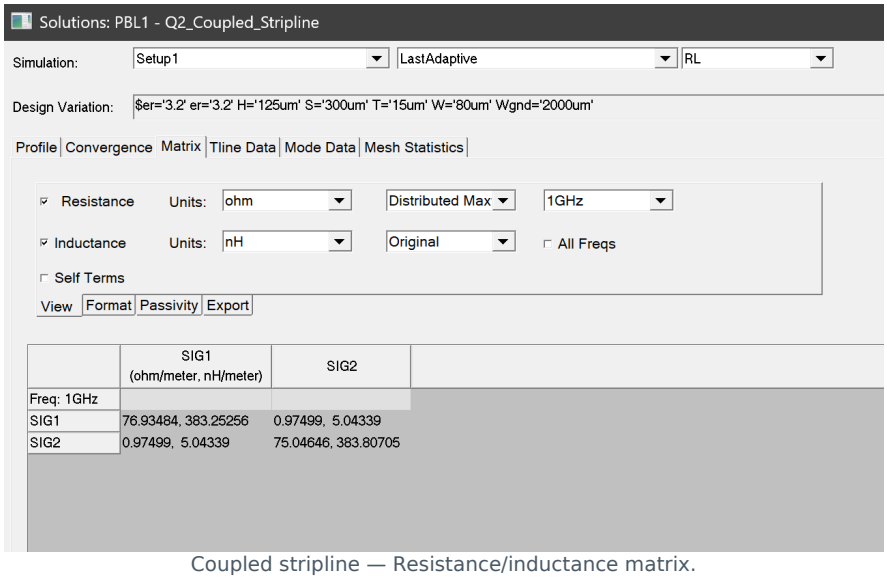
Coupled microstrip — Modal attenuation and velocity data.



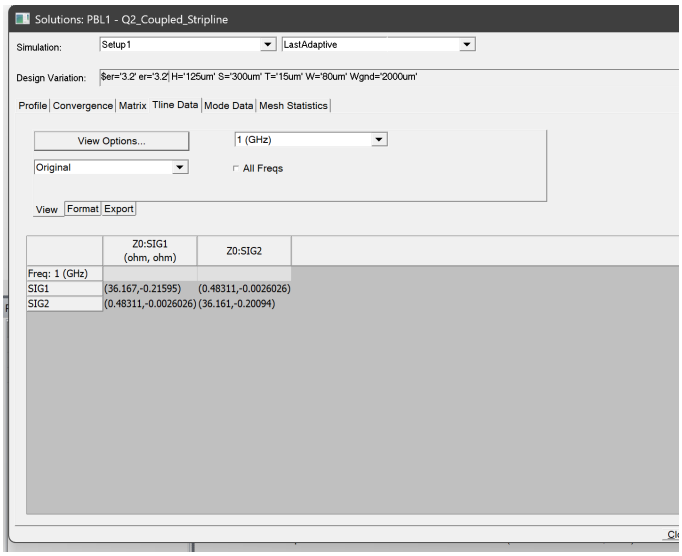
Coupled stripline geometry.



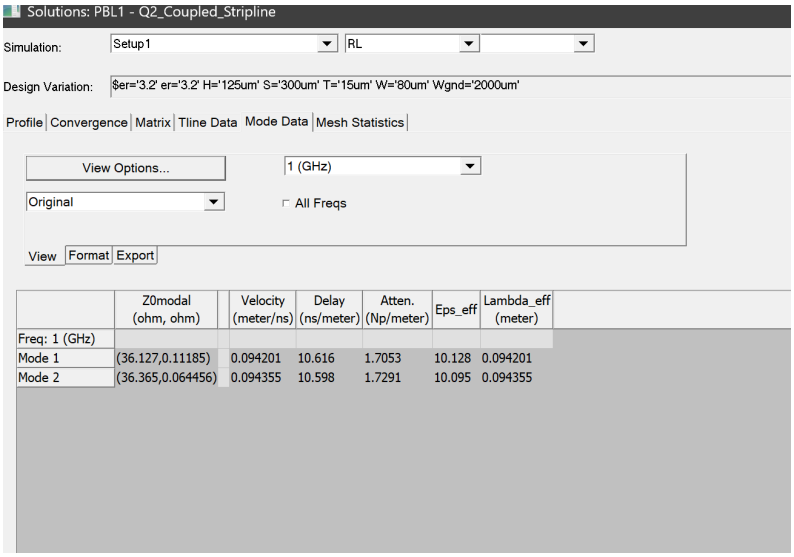
Coupled stripline — Capacitance/conductance matrix.



Coupled stripline — Resistance/inductance matrix.



Coupled stripline — Transmission-line impedance data.



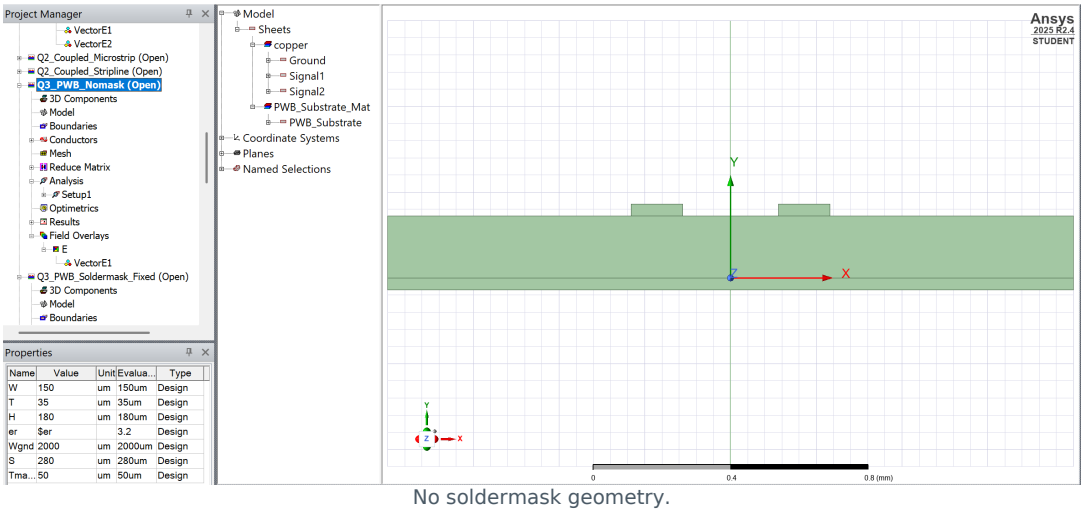
Coupled stripline — Modal attenuation and velocity data.

4. Q3 — PWB soldermask impact

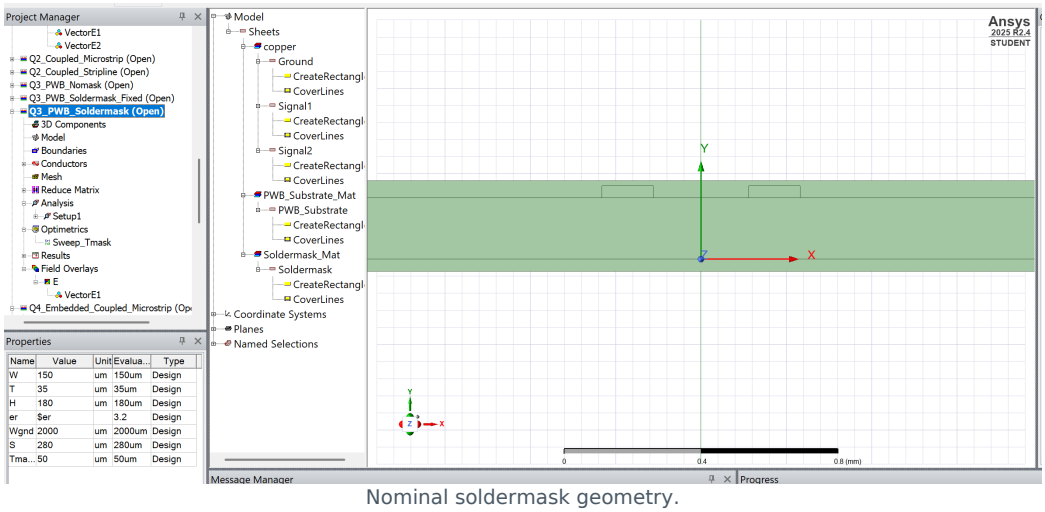
Two PWB microstrip traces were compared without soldermask and with a nominal 50 µm soldermask (εr = 3.5, tan δ = 0.02). Dielectric loading increased self and mutual capacitance and reduced characteristic impedance. For thicknesses below the 35 µm copper height, the original full-width mask geometry intersected the conductors; a corrected three-region side-mask geometry was used for 5–30 µm. Results for 35–60 µm were obtained using the original covered-trace model. The two regimes are not identical geometrical constructions and should be interpreted accordingly.

1 GHz quantity	No mask	50 µm mask	Change
C11 (pF/m)	81.89579	93.65127	+14.35%
|C12| (pF/m)	5.76878	9.18063	+59.15%
Re(Z0,11) (Ω)	71.215	66.768	−6.24%
Mode 1 attenuation (Np/m)	0.60275	0.69638	+15.53%

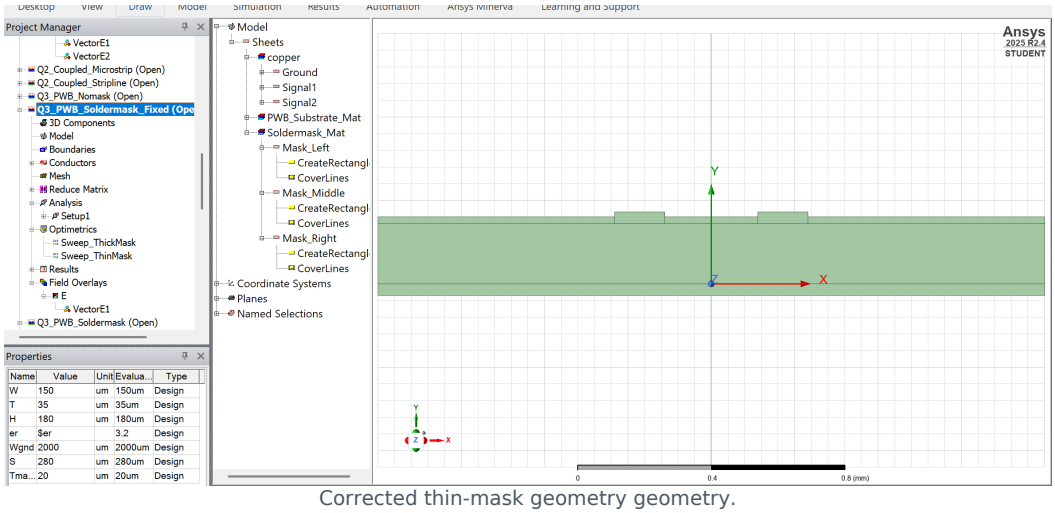
Soldermask replaces some surrounding air with higher-permittivity material, increasing electrostatic charge storage and coupling between traces. Increasing dielectric participation also raises the loss contribution of the covering material. The 0 µm result is taken from a separate no-mask model, rather than a degenerate zero-height sheet.



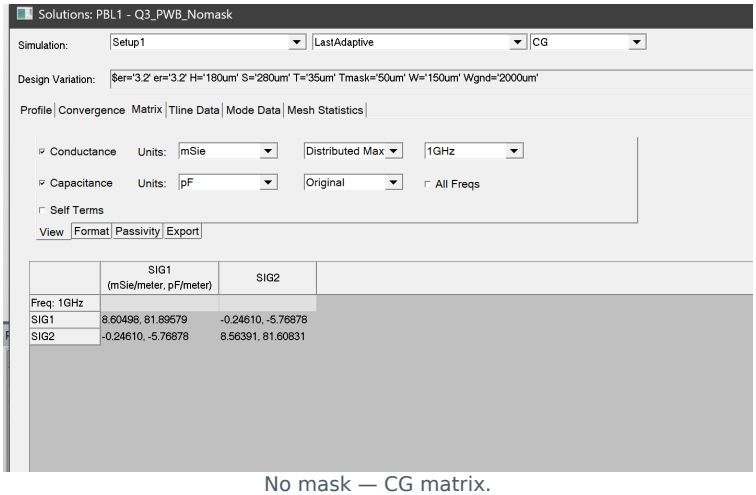
No soldermask geometry.



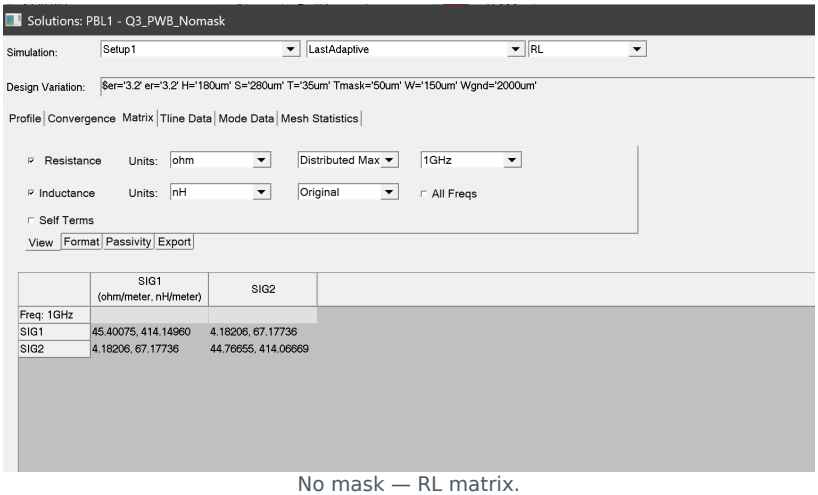
Nominal soldermask geometry.



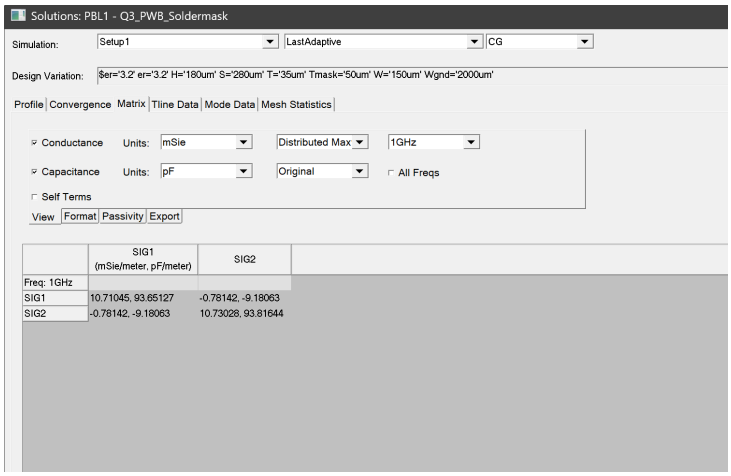
Corrected thin-mask geometry geometry.



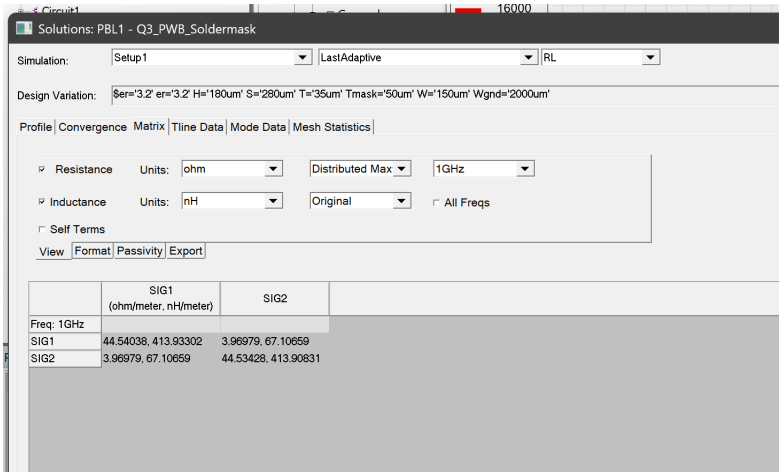
No mask — CG matrix.



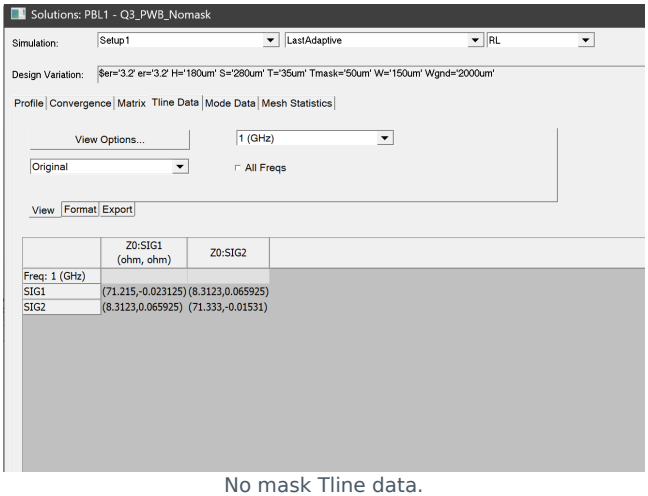
No mask — RL matrix.



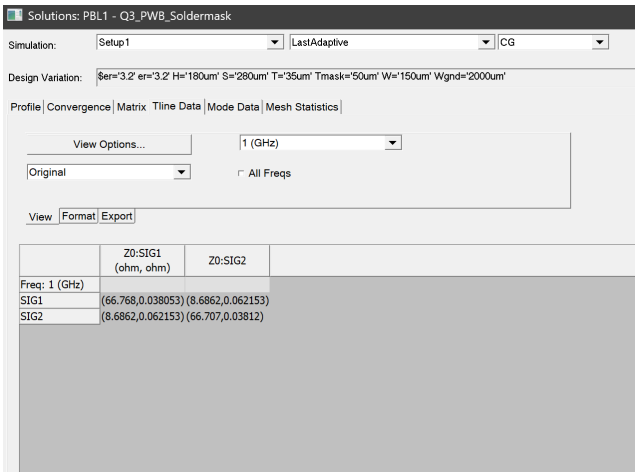
50 µm mask — CG matrix.



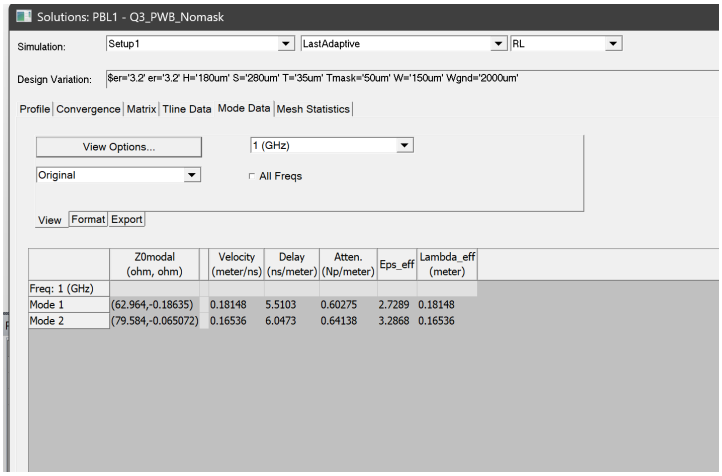
50 µm mask — RL matrix.



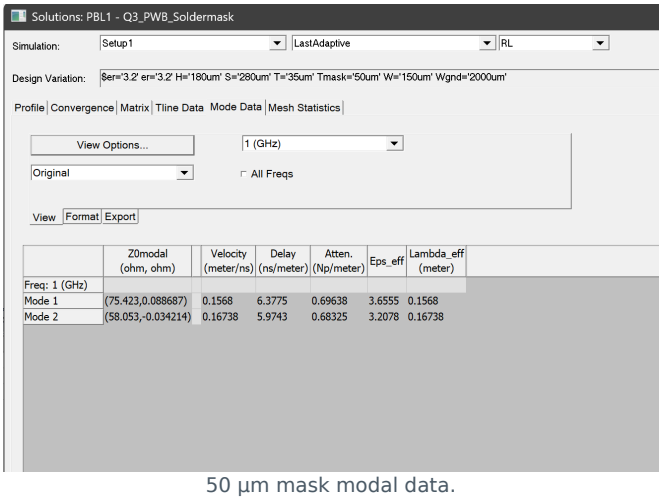
No mask Tline data.



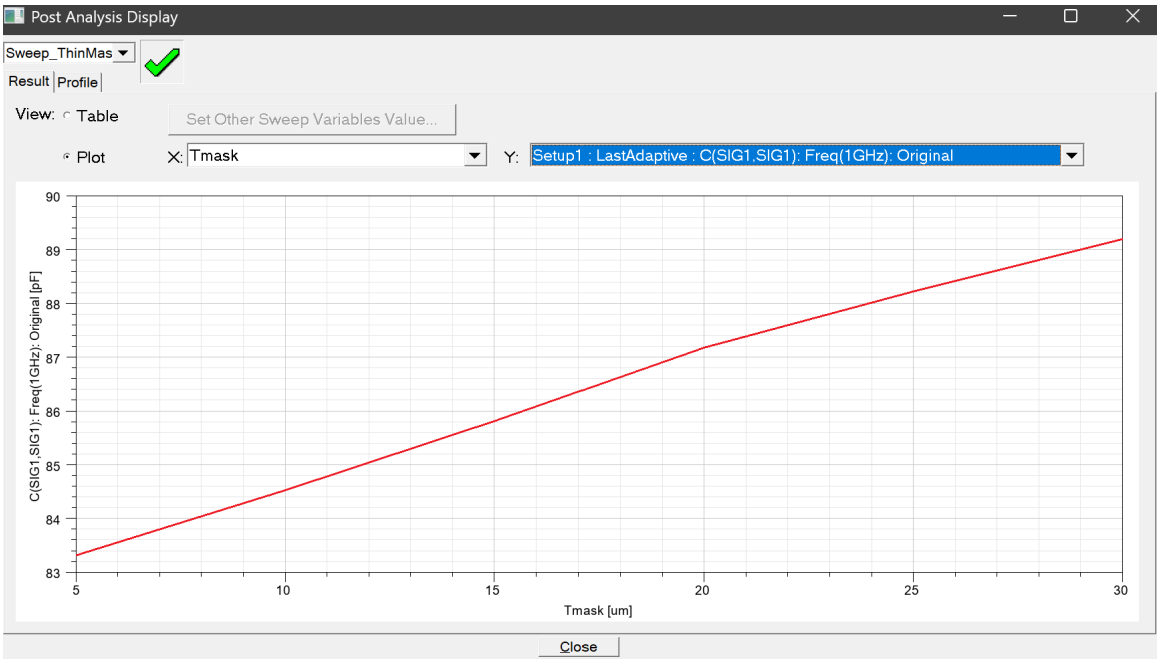
50 µm mask Tline data.



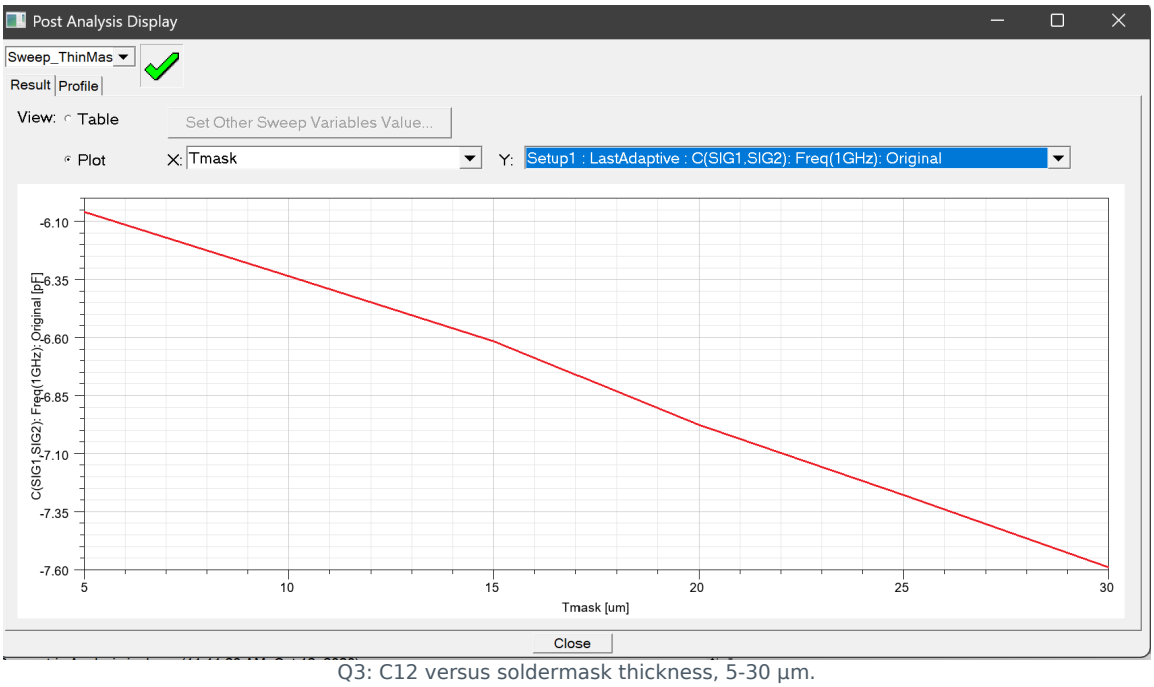
No mask modal data.



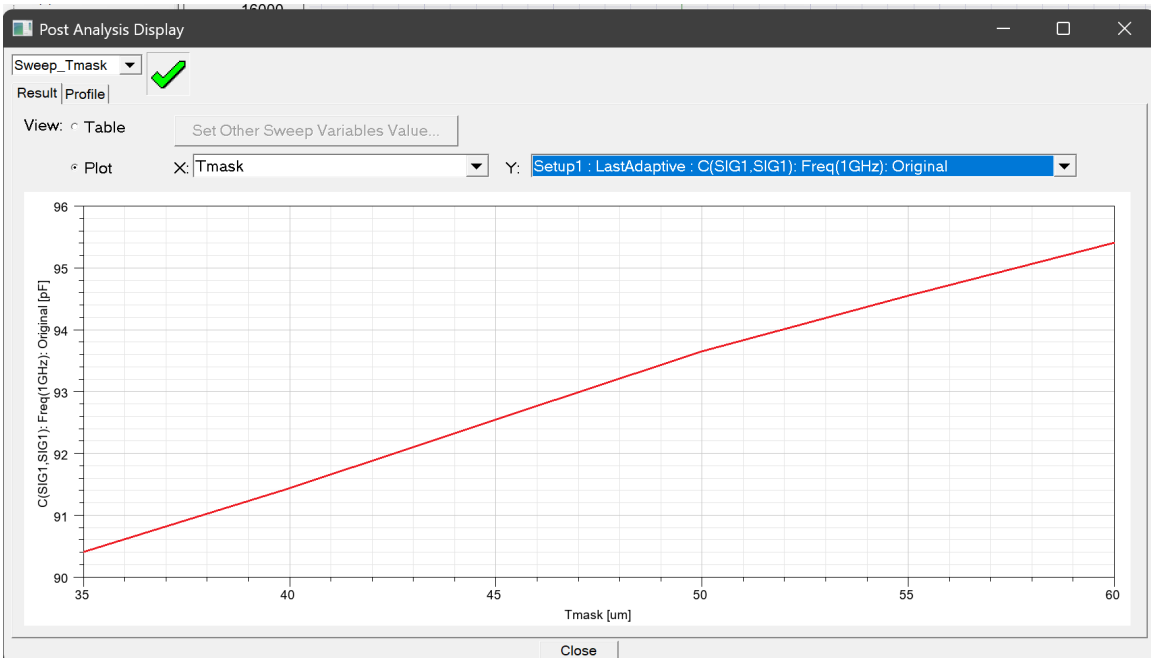
50 µm mask modal data.



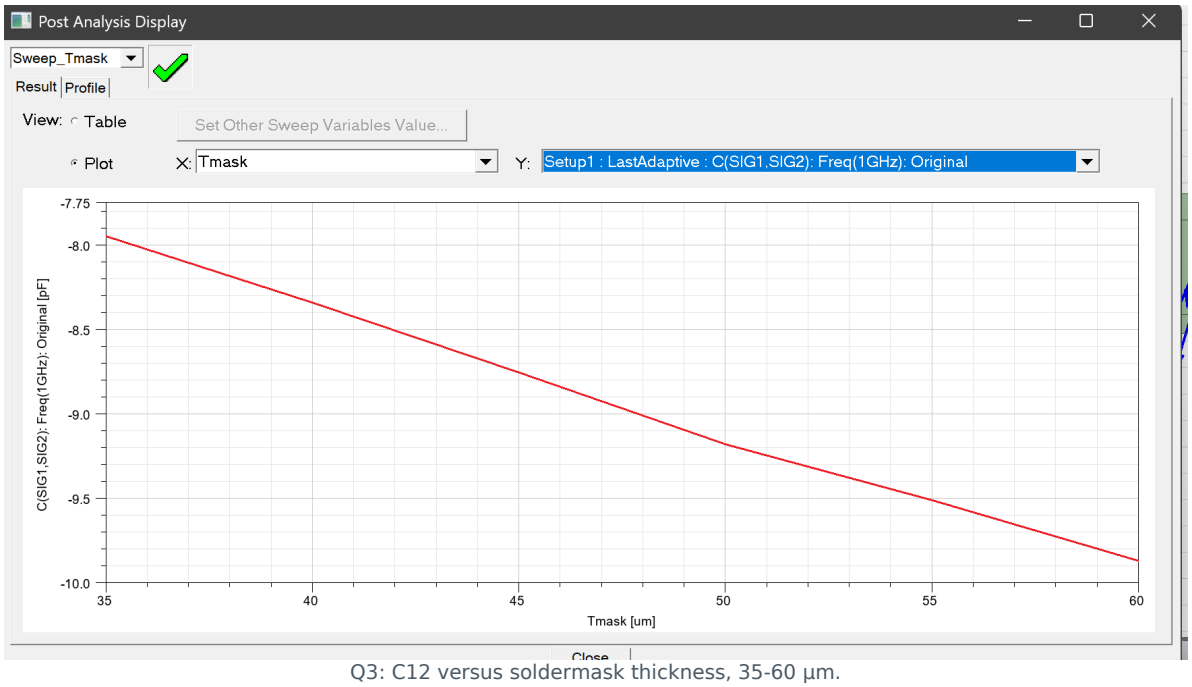
Q3: C11 versus soldermask thickness, 5-30 µm.



Q3: C12 versus soldermask thickness, 5-30 µm.



Q3: C11 versus soldermask thickness, 35-60 µm.



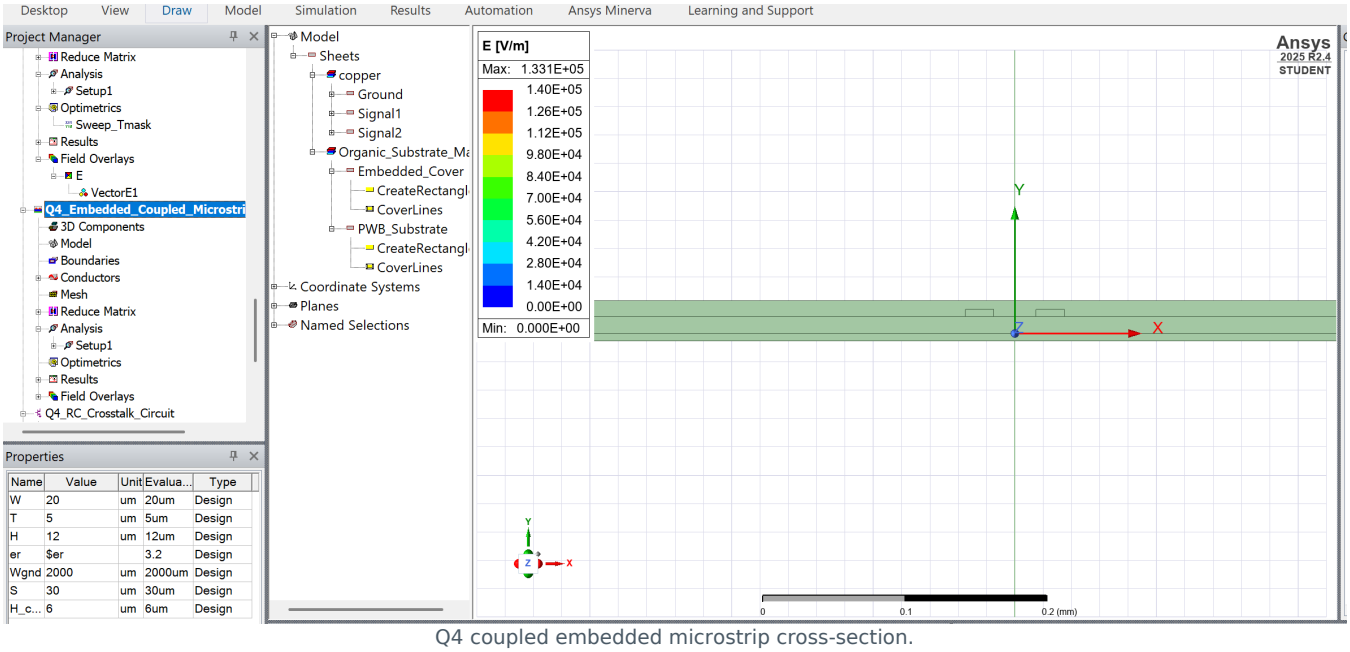
Q3: C12 versus soldermask thickness, 35-60 µm.

5. Q4 — Embedded organic coupled-line RC simulation

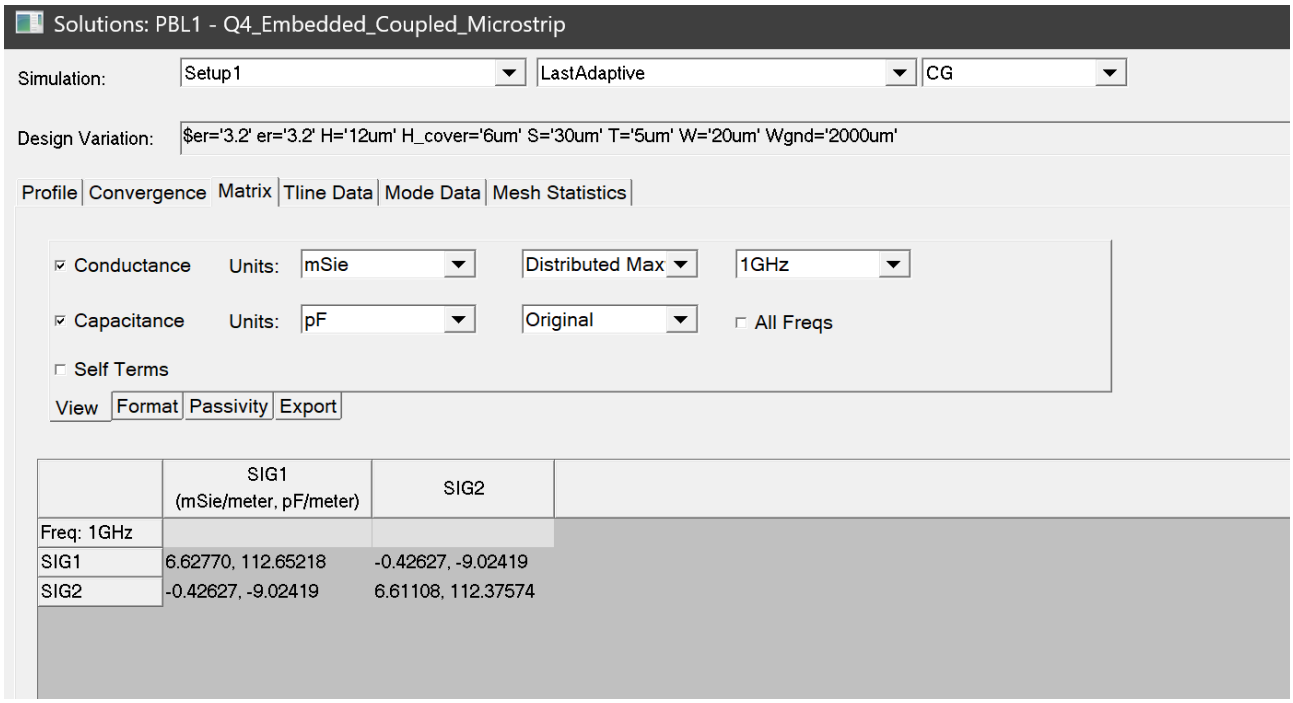
The Organic/Thin Film coupled embedded microstrip used W = 20 µm, S = 30 µm, T = 5 µm, H = 12 µm, and an assumed 6 µm dielectric cover. The 1 GHz per-unit-length matrices were converted to a 15 mm lumped RC circuit. A 40 Ω active transmitter, a grounded 40 Ω quiet-line driver, and 0.5 pF receiver capacitance on each line were used. The input pulse was 0–1.2 V, with 0.5 ns delay, 0.15 ns rise/fall time, 1.5 ns pulse width and 3 ns period.

Matrix / equivalent parameter	Value
R11 / R22 (Ω/m)	344.99263 / 344.79563
R12 (Ω/m)	29.53329
C11 / C22 (pF/m)	112.65218 / 112.37574
C12 (pF/m)	−9.02419
Line R1 / R2 over 15 mm	5.175 Ω / 5.172 Ω
Coupling capacitor Cm	0.13536 pF
Shunt Cg1 / Cg2	1.55442 pF / 1.55027 pF

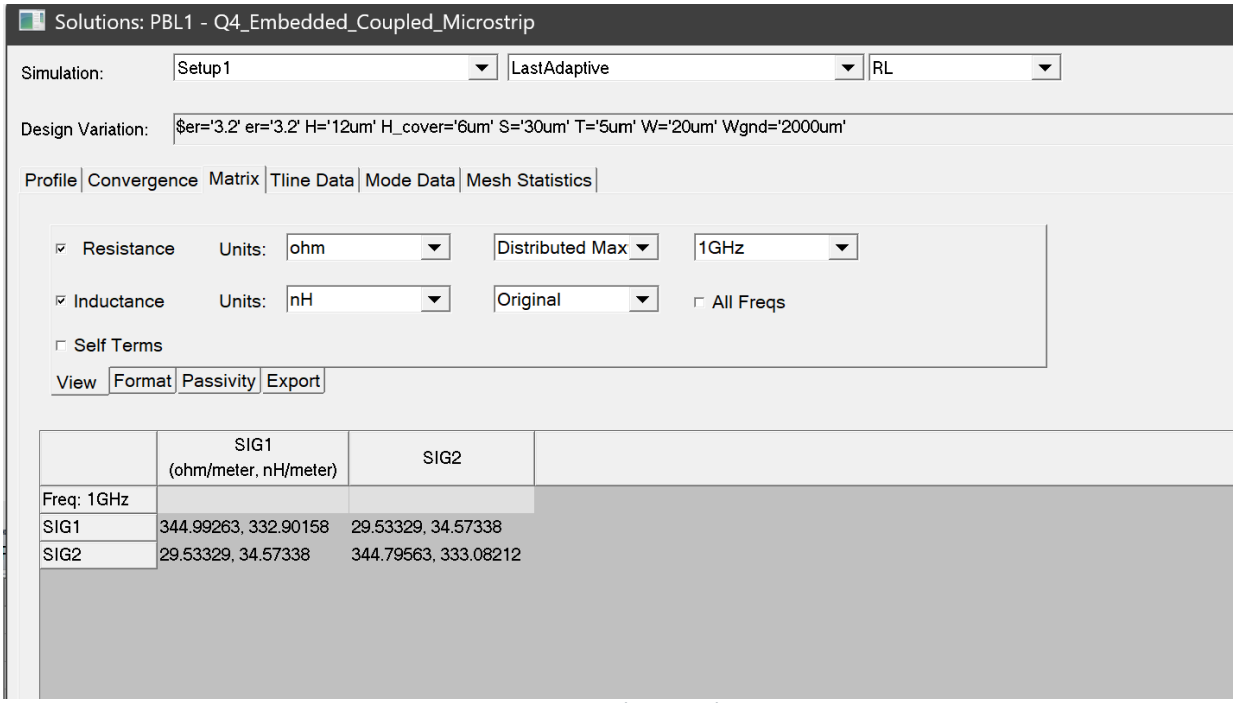
The ground shunt capacitors are derived from the Maxwell capacitance matrix by subtracting the mutual capacitor contribution: Cg1 = (C11 − |C12|)l and Cg2 = (C22 − |C12|)l. The circuit uses self-resistance terms; mutual resistance and inductive coupling are not explicitly represented in this simplified lumped RC approximation.



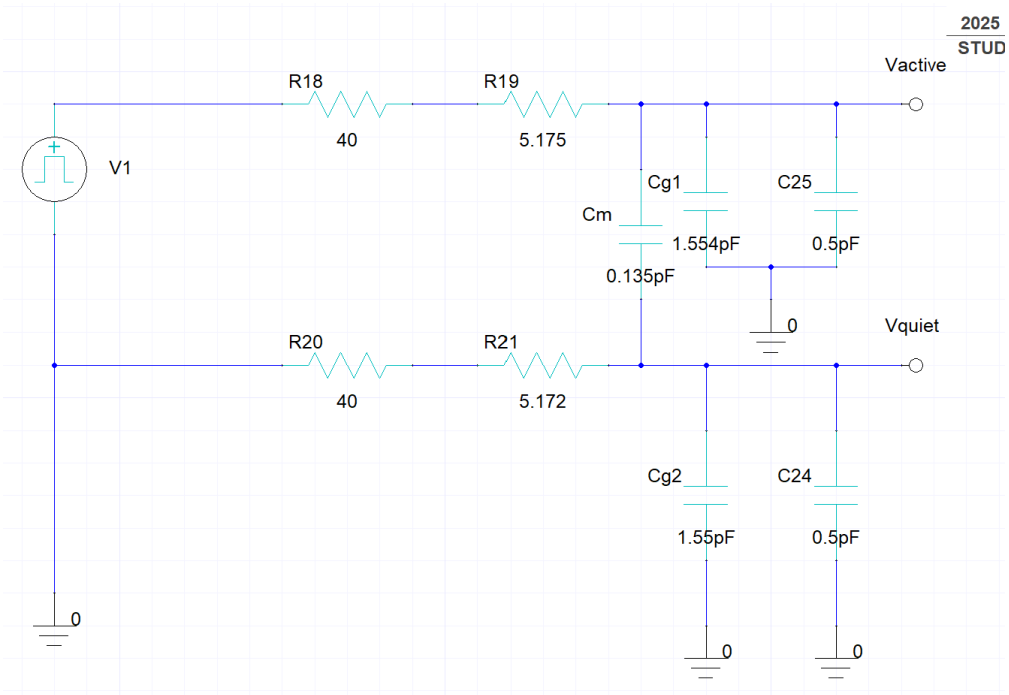
Q4 coupled embedded microstrip cross-section.



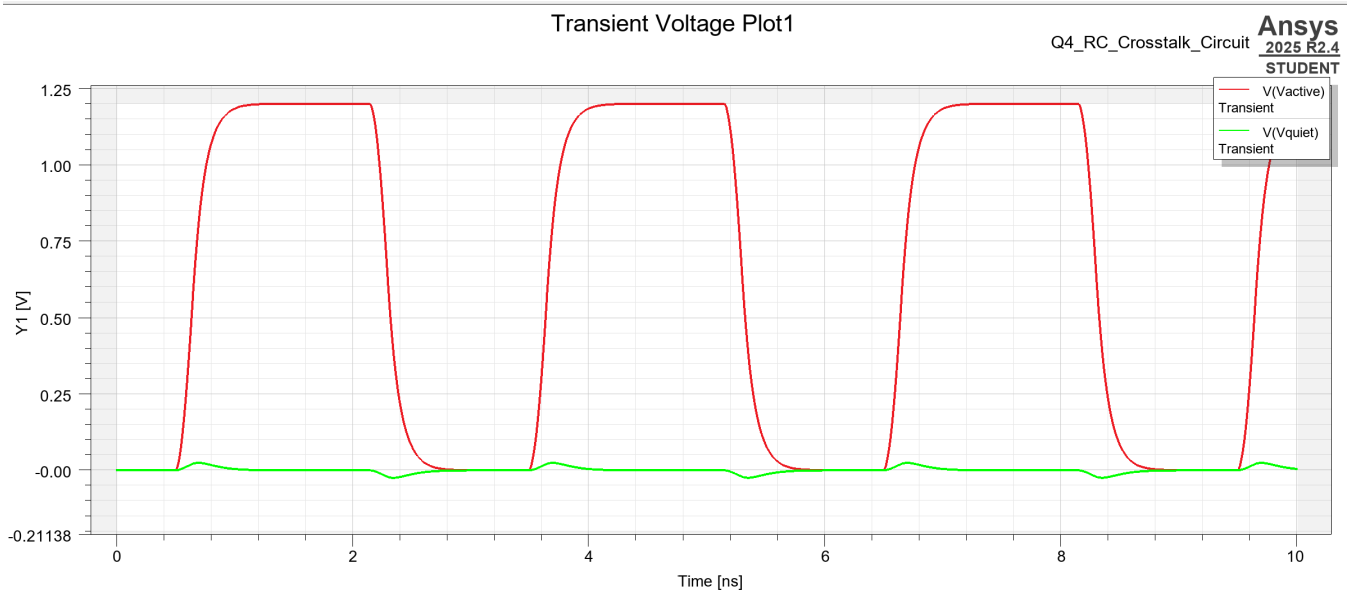
Q4 extracted CG matrix.



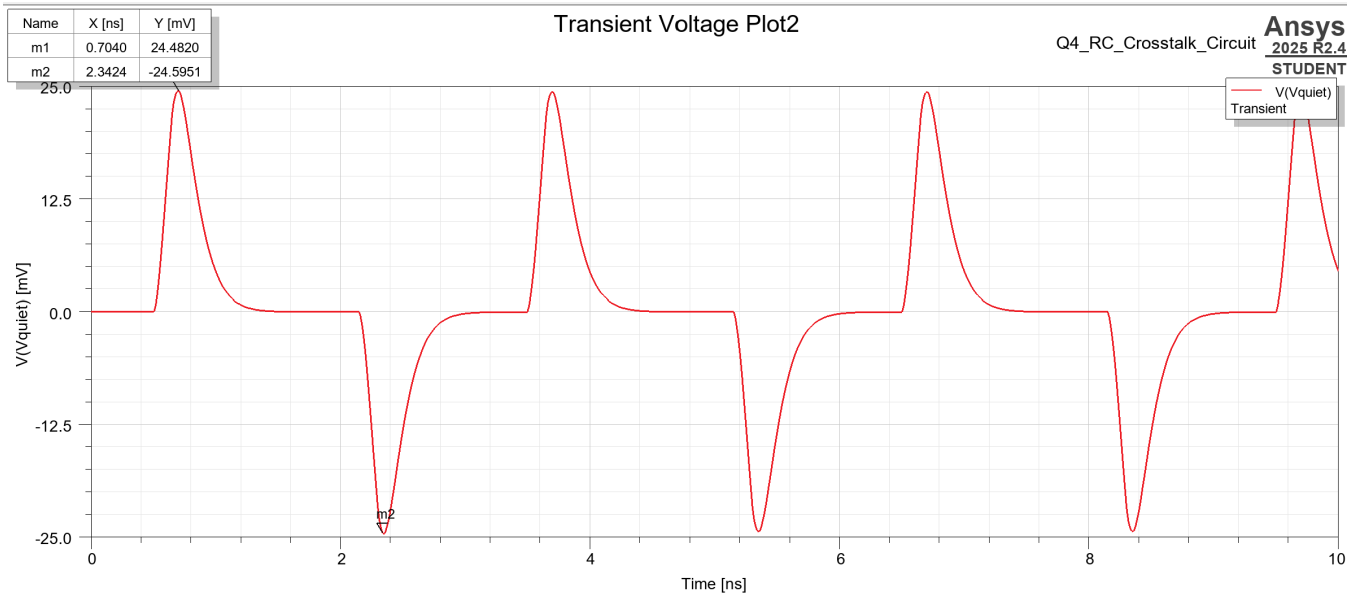
Q4 extracted RL matrix.



Ansys Circuit lumped RC schematic.



Active and quiet receiver voltage waveforms.



Measured quiet-line positive and negative crosstalk peaks.

Transient metrics from exported waveform data

Metric	Result / calculation
Input 50% crossing	0.575000 ns (ideal specified pulse)
Active output 50% crossing	0.653193 ns
50% propagation delay	78.19 ps
Active 10% / 90% crossings	0.559813 / 0.812655 ns
10–90% rise time	252.84 ps
Peak absolute quiet-line voltage	24.5951 mV
Peak crosstalk relative to 1.2 V	2.0496%

The 50% delay and rise-time thresholds were obtained by linear interpolation between adjacent exported samples. Peak crosstalk is the largest absolute quiet-line sample over the exported 0–10 ns interval. The delay is referenced to the ideal specified pulse source, not an independently exported input-node trace.

6. Q5 — Design recommendation to the system architect

Recommendation: For the modeled high-speed package bus, Organic/Thin Film with embedded microstrip routing is a reasonable candidate, subject to further distributed-line and noise-margin validation. Its 20 µm conductor width and 50 µm pitch provide fine routing density, while the Q4 lumped model yielded a 78.19 ps 50% delay, 252.84 ps rise time, and 24.595 mV peak quiet-line crosstalk (2.05% of the 1.2 V signal).

The ceramic comparison demonstrates a useful trade-off. At 1 GHz, coupled ceramic microstrip had lower modal attenuation (10.28 and 9.92 dB/m) than ceramic stripline (14.81 and 15.02 dB/m). Conversely, the stripline had lower normalized capacitive coupling (1.36% versus 1.91%), despite a higher absolute mutual capacitance. The PWB soldermask study further showed that dielectric coverage can increase both self and mutual capacitance and reduce impedance.

Therefore, selection depends on the system priority: use the simulated ceramic microstrip where attenuation dominates, consider stripline for lower normalized electrostatic coupling, and retain the organic embedded option where routing density and the demonstrated lumped RC behavior are primary considerations. Because the substrate comparisons use different dimensions and terminations, the present simulations do not prove that any one technology is globally optimal. A production decision would require matched-condition comparisons, frequency-dependent RLGC extraction, distributed interconnect modeling, and receiver noise-margin analysis.

7. Conclusions and limitations

The Q1 sweeps demonstrated the expected dependence of capacitance on conductor width, dielectric spacing and permittivity. Q2 quantified the distinction between absolute mutual capacitance, normalized coupling and modal attenuation. Q3 showed substantial dielectric-loading effects from soldermask and required two geometrical regimes for the thickness sweep. Q4 connected extracted physical parasitics to circuit-level propagation delay, edge degradation and capacitive crosstalk.

Limitations: Q1 precise ground-plane thresholds were not independently established and E-field screenshots were omitted by scope choice; this does not fully satisfy those portions of the assignment. Some Q3 RL sweep points exceeded the AEDT Student mesh limit, although reported baseline matrices and CG sweep curves were retained. The Q4 model is a lumped RC approximation, not a full distributed coupled transmission-line simulation.

References

1. Dr. Prashant Kumar, ECEG4071 IC Fab and Packaging — PBL 1: Interconnect Resistance, Capacitance & RC Behavior, supplied course brief.

2. Ansys Electronics Desktop Student, user-generated Q2D extraction matrices, Optimetrics plots, and Circuit transient data (12 October 2026).

AI Use Disclosure

ChatGPT (OpenAI) was used to understand and troubleshoot Ansys Q2D/Circuit setup, conductor assignments, Maxwell capacitance matrices, convergence and solver errors, lumped RC modeling, and transient waveform measurements. It also assisted in organizing simulation evidence and technical explanations. The AEDT simulations and screenshots were generated by the student.